

Machine Learning tools Comparative Analysis assignment - SBA25255

Organization Description

Stripe is a 'fintech' company which over the years has evolved from being a payments provider to offering a full blown financial infrastructure. This provides customers with what Stripe calls "revenue and finance automation". Their financial infrastructure manages a full range of services including payments, billing, tax, fraud management, card issuance, identity management and much more. Stripe's mission is to "Grow the GDP of the internet"[1]

Some interesting facts about Stripe -

1. 1.5% of global GDP runs on Stripe[2]
2. Stripe support 135+ payment methods including fiat currencies and crypto stablecoins[2]
3. Stripe processed \$1.4tn worth of payments in 2024[2]
4. Stripe have 200m+ active billing subscriptions[2]
5. 4.7m websites worldwide use Stripe[5]

Given the above scale AI is critical to both Stripe's and their customers ongoing success and growth.

Stripe's AI use case for this assignment

Given the above scale Stripe has an automated onboarding process for new customers. Part of this process involves 'Know your Customer' (KYC)[3] anti money laundering (AML)[4] regulatory requirements.

As part of onboarding the customer is initially asked to key in some specific information about their business. The KYC regulatory requirements for the geographical region the business is located in are then executed.

KYC has potentially two steps.

- (1) The first step is for the customer to "key in" information about themselves. Like their name / address / date of birth. Checks are then run on this information to validate they are who they say they are. This step takes seconds.
- (2) However if the identity verification fails then the customer is moved to the second step in the KYC process. The customer is requested to upload identity documentation. This step in the KYC process can take hours to complete. It involves a human manually inspecting uploaded documents vs the details the customer entered. Then making a determination as to whether
 - (a) The identification document provided is valid
 - (b) The customer is who they say they are. The details of each identity document review are recorded in an online system.

Attributes of the problem to be solved

- When a customer submits an identity document we want to **predict** -
 - The document itself is valid
 - The information entered by the customer matches what's on the identification document
- The above requires a series of **decisions** to be made. For example -
 - Does the format of the document match how it should appear? For example if it's a french drivers licence the expiry date should be on the front on line 4b
 - Is the expiry date < today ?
 - Does the information entered by the customer match what's on the document
- Stripe has a system that records details of all of the manual reviews. I.e. we have historical **labelled** data.
- Stripe has had year on year (yoy) growth of 38%[4] so we can assume this is a **high volume / large data set environment**.

Achieving the above solution will benefit Stripe in the following ways -

1. Reducing step 2 in the KY process from hours to seconds which in turn will remove significant friction for new customer onboardings.
2. Ensure that this step in the KY process does not become a barrier to Stripe's continued growth rate

Potential Approaches

Unsupervised or supervised learning?

We want to predict upon document submission that the document itself is valid and that the customer is who they say they are. We want to leverage historical labelled data from previous document reviews. Supervised learning is strongly aligned with our use case.

Unsupervised learning is best used for gaining insights into patterns and structure of data where a “correct answer” is not required.

Supervised learning is strongly aligned with our use case.

Classification or clustering ?

Clustering is grouping data points into clusters and then reviewing the clusters to gain insights. It's pure data discovery where there is no explicit answer or prediction.

Classification involves providing an explicit prediction. We can use Stripe's historical document review labeled data as an input to making the prediction. Classification can predict either a category such as determining whether a given email is spam. Or predict a continuous value such as what the outside temperature will be tomorrow. Classification works well against very large data sets such as this Stripe use case.

Classification is strongly aligned with our use case.

Algorithm options

K-Means

This algorithm is for unsupervised learning. It clusters data when in turn provides insights on data structure and patterns. It predicts a label of a data point from labels of closest neighbours. There's no training required and it works well in small data sets.

Given we will use supervised learning, K-Means is not applicable for our solution. Also the Stripe use case involves large datasets which is not suitable for K-Means.

Naive Bayes

This algorithm is for supervised learning and has the following attributes -

- Assumes features are independent of each other
- Used as a classifier for predicting categories
- Fast and scalable

All of the above fit the Stripe use case. However the Stripe use case will involve decision making. For example if a customer submits a French drivers licence here are some of the decisions we will need to make -

- Is the customer's business located in France? If not then we fail
- Is the expiry date < today ? If not then we fail
- Is the expiry date not on the front at line 4b ? If not then we fail

Naive Bayes does not align with the Stripe use case decision making needs.

Decision Trees

This algorithm is for supervised learning and has the following attributes -

- Prediction of categories or a continuous number.
- Commonly used for approval processing and fraud detection
- Training involves asking a series of questions on labelled input data
- Has proven to be successful with very large datasets[6]

The decision tree algorithm is best aligned to the Stripe use case

Recommendation

Use supervised learning with a decision tree algorithm for the Stripe use case.

References

- [1] <https://stripe.com/blog/untapped-opportunity-at-the-bottom-of-your-customer-funnel> - Retrieved 2026-02-19
- [2] <https://stripe.com/ie> - Retrieved 2026-02-19
- [3] <https://www.redflagalert.com/articles/a-guide-to-kyc-regulations-in-2024> - Retrieved 2026-02-19
- [4] <https://www.centralbank.ie/regulation/anti-money-laundering-and-countering-the-financing-of-terrorism/regulatory-requirements-guidance> - Retrieved 2026-02-19
- [5] <https://www.chargeflow.io/blog/stripe-statistics> - Retrieved 2026-02-19
- [6] <chrome-extension://efaidnbmnnnibpcajpcgclefindmkaj/https://cdn.aaai.org/KDD/1998/KDD98-001.pdf> - Retrieved 2026-02-19